

A Multi-Cohort Decision-Evaluation Benchmark for EHR World-Model Components and Offline Reinforcement Learning

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ABSTRACT

Electronic health record (EHR) world-model components are increasingly connected to offline reinforcement learning (RL), yet comparisons can be driven by incompatible tasks, actions, rewards, support rules, or evaluator assumptions rather than algorithmic differences. We introduce a development benchmark that standardizes these layers across six scale-qualified ICU cohort targets and compares six transition methods, five uncertainty-capable methods, 34 policy/planner labels, and nine off-policy evaluation (OPE) estimators. We define a minimum decision model by transition (P), reward (R), and termination (T) components, while explicitly excluding an unmodeled observation process from the claim. On the three unchanged cohort lineages with complete current results, the Transformer leads one-step error, the Gaussian ensemble leads recursive error and sharpness, and GRU-D leads recursive calibration. In matched exact-finite mechanisms, model-free and world-model planners trade places and no learned method is uniformly best. Across repeated logged datasets for four heterogeneous mechanisms, no frozen OPE contract passes all gates, so we report no retrospective EHR policy value or winner. Sepsis, AF/flutter, and heart-failure results from smaller superseded constructions are excluded pending preregistered reruns on their restored large lineages. The contribution is an executable multi-cohort scaffold, a complete all-method inventory, and a failure decomposition from component forecasting through policy-evaluation authorization.

CCS CONCEPTS

• Applied computing → Health informatics; • Computing methodologies → Reinforcement learning.

KEYWORDS

electronic health records, world models, offline reinforcement learning, benchmark, off-policy evaluation, uncertainty

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1 INTRODUCTION

EHR prediction benchmarks standardize cohorts, preprocessing, splits, and metrics [9, 28, 29]. Offline RL over EHR trajectories adds a more fragile technical chain: recorded state must precede the action label, reward must follow it without mechanically encoding it, policy probability must remain inside empirical support, learned dynamics must remain stable under recursion, and OPE must recover known values before retrospective scoring is authorized [24,

25]. Sepsis RL established a widely reused fluid-by-vasopressor action abstraction [13]; recent work connects latent EHR dynamics and imagination to policy learning [30]. These studies motivate a shared evaluator, but their different task and scoring contracts prevent direct algorithmic comparison.

For clarity, our minimum decision model is

$$\hat{M} = (\hat{P}, \hat{R}, \hat{T}), \quad (1)$$

where $\hat{P}$ models the action-conditioned next state, $\hat{R}$ supplies the benchmark objective or reward-relevant outcome, and $\hat{T}$ models termination or censoring. A transition predictor alone is therefore a P component, not model-based RL: planning requires both an objective and a return horizon [23]. We call these $P/R/T$ world-model components, rather than a complete EHR generator, because the future observation process O is not generated.

Our primary contribution is the standardized comparison and failure decomposition across $P/R/T$ component learning, component model $\times$ planner combinations, model-based and model-free offline RL, known-value truth, retrospective support diagnostics, and cross-cohort evaluability. Transition accuracy is a validity layer rather than the endpoint. Temporal ordering and leakage control are methods requirements rather than the headline contribution.

Because no genuinely untouched role remains, all EHR findings are labeled *development benchmark evidence*. The benchmark evaluates $P/R/T$ component and model-free offline-RL pipelines across heterogeneous EHR tasks, compares policy behavior against exact truth in constructed response mechanisms, and audits support, collapse, planning stability, and model exploitation on retrospective EHR data. The retrospective layer evaluates computational behavior, not policy value.

Contributions. We provide: (1) a common, time-ordered task scaffold with train-only fitting and subject-disjoint development roles; (2) an AI-Clinician-aligned K25 sepsis scaffold plus five task-specific extensions; (3) complete all-method transition, uncertainty, policy, and OPE inventories rather than winner-only summaries; (4) a heterogeneous exact-finite stress family that separates fixed-extreme dominance from adaptive policy discrimination; (5) a component-level bridge from prediction and uncertainty to known-value policy behavior and model exploitation; (6) repeated-dataset OPE coverage and authorization rather than estimator names or single-dataset interval inclusion; and (7) explicit non-execution when probability, support, or evaluator prerequisites fail.

Research questions. The benchmark asks: **RQ1**, how do $P/R/T$ component families differ in one-step, recursive, reward, termination, uncertainty, and action-information fidelity? **RQ2**,

how do model-based and model-free policies behave under interior-optimum, state-dependent, heterogeneous-response, and delayed-trade-off mechanisms? **RQ3**, when do learned models exhibit exploitation gaps, and which OPE contracts recover known policy values and ranks under repeated logged-data sampling? **RQ4**, which parts of this decision scaffold remain operational across heterogeneous cohorts without transferring sepsis-specific reward semantics?

2 RELATED DECISION BENCHMARKS

Table 1 separates task-specific retrospective EHR RL, EHR component/simulation models, and synthetic known-value benchmarks. It is representative rather than systematic: the purpose is to expose which evidence layers are connected under one contract, not to claim exhaustive coverage.

Prior work occupies distinct evidence surfaces. Real-EHR studies usually use task-specific retrospective evaluators; MedDreamer applies one method scaffold to two task-specific contracts; general EHR models may stop at trajectory simulation; and synthetic benchmarks supply known returns without validating real-EHR task construction. Sepsis-specific MDP and trajectory resources improve reproducibility within that domain but do not supply a shared multi-task world-model-policy-evaluator contract [2, 10]. We connect six real-EHR task contracts to common P/R/T diagnostics, matched policy inventories, known-value evaluator tests, and explicit retrospective authorization boundaries. Our respiratory and shock tasks remain published-inspired adapters rather than reproductions of DeepVent or POPCORN. Healthcare offline-RL model selection remains sensitive to reward, support, and estimator choice [17, 24].

3 BENCHMARK SCAFFOLD

3.1 State, action, target, and roles

Each transition uses a fixed four-hour clock. Let

$$x_t = [v_t, m_t, \delta_t, c, a_{t-1}], \quad y_t = v_{t+1}, \quad (2)$$

where v_t is the pre-action value vector, m_t its observation mask, δ_t feature-wise recency, c static/pre-anchor context, and a_{t-1} the previous action abstraction. Current action, post-action measurements, future masks, post-outcome fields, and whole-stay diagnosis proxies are prohibited from x_t . The logged action a_t summarizes recorded exposure or settings over $[t, t + 4h)$; y_t begins at $t + 4h$. Every evaluated bin remains inside one ICU stay.

Throughout, $K = |\mathcal{A}|$ denotes the number of discrete action classes, not the number of observed transitions or treatment dimensions.

The interval label is deliberately called a *recorded-exposure action*: it can include adaptation during the interval and is not asserted to be fully assignable at its start. It is valid for logged-action-conditioned forecasting and retrospective policy-behavior diagnostics. In the known-value environments, by contrast, a_t is an assignable intervention in the constructed transition kernel. This distinction prevents the real-EHR label from inheriting the stronger semantics of the simulator action.

Imputation, normalization, action cutpoints, support masks, feature selection, behavior denominators, hyperparameters, and

checkpoints are fitted or selected on development train/validation roles only. The roles are subject disjoint, but historical use of the available evaluation roles precludes calling any role a new confirmatory holdout. No row is used to claim a confirmatory best method.

3.2 AI-Clinician-aligned K25 sepsis scaffold

Sepsis is aligned to the AI Clinician lineage only at the four-hour fluid-quintile $\times$ vasopressor-quintile decision scaffold ($K = 25$). It uses value, mask, recency, and prior-action views with locally re-constructed SOFA/lactate reward components. It is not an exact reproduction: numerical reward coefficients and parts of the source cohort implementation are not verifiable from the current artifact; the stay-bounding rules differ; the local SOFA proxy can mechanically overlap vasopressor action; and anchor-relative 90-day outcome ascertainment is unavailable. We therefore retain an aligned scaffold and treatment-isolated reward sensitivity, not a canonical reproduction. The final target is the broader 27,236-episode, 22,437-subject K25 lineage. Results from the 3,440-episode strict construction are superseded and excluded from primary synthesis because the two constructions are not scientifically interchangeable.

The matched feature/reward/denominator comparison did not support choosing a preferred setting because terminal outcome availability and matched semi-synthetic reliability were incomplete. The K25 scaffold remains a compatibility surface despite weak factual action information.

3.3 Task-specific extensions

Table 2 freezes the final cohort-scale targets. An episode is one eligible ICU-stay trajectory; subjects and episodes are both reported because repeated stays make them non-interchangeable. Primary eligibility requires at least 10,000 subjects *and* 10,000 episodes. Episodes should not be summed across cohorts because one person or stay may satisfy more than one task definition. Changing from a superseded small construction to a large lineage invalidates its old performance numbers; the affected cohort is shown as rerun-required rather than populated with transferred metrics. Respiratory uses observed PEEP quintile $\times$ observed FiO₂ quintile ($K = 25$); a row is excluded when either setting is unavailable rather than mapped to no action. Its primary dense benchmark reward is

$$r_t^{\text{resp}} = \mathbb{I}[94 \leq \text{SpO}_{2,t+1} \leq 98] - .5\mathbb{I}[\text{SpO}_2 \text{ outside}] + \mathbb{I}[70 \leq \text{MBP}_{t+1} \leq 80] - .5\mathbb{I}[\text{MBP outside}], \quad (3)$$

with an SpO₂-only sensitivity. Shock uses fluid quintile $\times$ vasopressor quintile and two separated reward components,

$$r_t^{\text{MAP}} = \text{clip}((\text{MBP}_{t+1} - 65)/25, -1, 1), \quad (4)$$

$$r_t^{\text{lac}} = -\tanh(\text{lac}_{t+1} - \text{lac}_t).$$

All are frozen post-action benchmark objectives rather than transferred sepsis semantics. Sepsis world-model reward diagnostics use the lactate component; its reference terminal outcome is unavailable. AF/flutter remains world-model-only because strict action-reward ordering is unresolved. AKI and heart failure retain their original P/T transition surfaces and add separate

Table 1: Representative landscape of EHR decision-model evaluation. “Known return” means value supplied by the environment rather than estimated from retrospective trajectories. Multisite evaluation is separated from multiple decision-task contracts.

Study	Data and decision task	Model / policy layer	Evaluation evidence	Cross-task scope
AI Clinician [13]	MIMIC-III + eRI; sepsis fluid-vasopressor decisions	Finite MDP + policy iteration	Retrospective OPE and action–outcome concordance; no known return	One task; two datasets
POPCORN [4]	MIMIC-III acute hypotension; hourly K20 fluid–vasopressor actions	Prediction-constrained POMDP + PBVI	Forecasting/likelihood + CWPDIS, ESS, and support; no known return	One task
DeepVent [14]	MIMIC-III; four-hour ventilation settings	Model-free CQL	FQE and action-distribution diagnostics; no known return	One task
MedDreamer [30]	MIMIC sepsis + eICU ventilation	Latent RSSM + imagination-based policy learning	Learned-reward/relative-mortality metrics + CWPDIS/ESS; no known return	Two tasks; task-specific actions/rewards
EHR-MPC [21] (preprint)	Eight-hospital ICU data; sepsis policy evaluation	Generative EHR twin + MPC	Importance weighting + learned-twin simulation; no known return	One policy task; multiple sites
EHRWorld [20] (preprint)	MIMIC-IV longitudinal trajectories	Intervention-aware long-horizon model; no policy learner	Forecasting and simulation fidelity	No policy benchmark
EpiCare / DTR-Bench [8, 18]	Synthetic clinical decision tasks	Multiple offline and online RL methods	Simulator true return + OPE	Multiple synthetic tasks; no real-EHR task construction
Ours	MIMIC-IV; six real-EHR task contracts	Four P/R/T component families + matched policy layers	Adaptive exact-finite truth + monotone sensitivity + retrospective authorization boundaries; no EHR winner	Shared cross-task contract

decision-task extensions: severe AKI uses daily wait versus initiate RRT ($K = 2$, absorbing after initiation, at most three decisions), while heart failure uses a six-hour loop-diuretic-quartile by vaso/inotrope indicator action ($K = 8$, at most 12 decisions). Their post-action reward proxies define task-matched known-value mechanisms; they do not create retrospective EHR policy values.

4 METHODS COMPARED

4.1 P/R/T component models and metrics

The evaluated component is

$$\hat{M} = (\hat{P}_\theta(v_{t+1} \mid x_t, a_t), \hat{R}, \hat{T}), \quad (5)$$

not a complete generator of the future observation process. $\hat{P}$ predicts next values, $\hat{R}$ is a frozen reward head or deterministic reward mapping, and $\hat{T}$ predicts termination/censoring. The four core families span missingness-aware recurrence (GRU-D) [1], causal attention (Transformer) [27], stochastic latent recurrence (categorical RSSM), and explicit predictive dispersion (Gaussian recurrent ensemble); persistence and histogram gradient boosting are lower-capacity controls. This is a component-family $\times$ planner factorial comparison, not a claim to exhaust every integrated model-based RL system. Neural runs use seeds 3408/3411/3414, hidden dimension 48, latent dimension 16 where applicable, batch size 512, learning rate 0.002, and weight decay 10^{-4} . Checkpoints minimize 0.7 RMSE + 0.3 MAE on development validation after train-only fitting, with 8 minimum and 40 maximum epochs, patience 6, and one uniform 80-epoch extension when the frozen convergence trigger fires. The Gaussian ensemble combines the three recurrent members and is one derived row per task.

One-step RMSE/MAE uses only observed target cells after train-fitted normalization. Recursive rollout starts from the first validation state, feeds each predicted value back, marks generated future values as unobserved, increments recency, and follows the logged action sequence for eleven four-hour steps. At each horizon, error is scored against cells observed at that step in the recorded target mask. The scalar recursive error is the normalized observed-cell RMSE at the final executed horizon (step 11, 44 hours), averaged

across available executed-seed rows for a method–cohort pair; all 924 horizon-resolved rows are released. Thus recursion tests value-state compounding under a fixed missingness convention; it does not claim to generate future measurement decisions. Reward heads are deterministic mappings from predicted state when available. Termination is evaluated natively, after train-only recalibration, and against a train empirical-hazard baseline.

Uncertainty reporting includes Gaussian NLL/CRPS, coverage at 50, 80, 90, and 95%, interval width and score, and risk–coverage [6, 15]. Factual action information uses one fixed primary estimand throughout: shuffled-action minus observed-action RMSE on reward-relevant targets within matched strata, averaged over the four core model families and executed seeds. Each model–seed contrast also carries a 500-replicate materialized-episode bootstrap interval. It is a predictive-information diagnostic.

4.2 Planners and policies

Planner labels encode how many future decision steps are optimized. **H1** exhaustively scores every supported current action under one predicted step. **H4** and **H8** optimize nominal four- and eight-step action sequences with categorical cross-entropy method (CEM) planning, execute only the first action, observe the next constructed state, and replan. Each CEM iteration evaluates 64 sequences, retains eight elites, updates categorical probabilities with smoothing 0.2, and applies the frozen support mask at every step. Three iterations and matched model-query budgets are used. The effective horizon is truncated by the remaining task horizon; therefore AKI’s nominal H4 and H8 variants are both H3-effective and are retained only as budget-contract diagnostics. “+U” subtracts 0.1 times predictive scale as an uncertainty penalty.

The model-free core includes behavior cloning, independent implementations of discrete BCQ [3] and CQL [16], a Soft-SPIBB conceptual adapter, and a Decision Transformer contract adapter. Five controls—behavior, supported random, minimum action, maximum action, and a state-dependent severity rule—remain separate from learned methods. Neural runs use three fixed seeds, one train/validation trial, at most 20,000 training windows, 12 epochs, hidden dimension 64, batch size 512, and discount 0.99.

Table 2: Frozen final cohort targets. Primary inclusion requires at least 10,000 subjects and 10,000 episodes; counts are aggregate eligible-lineage scale, not transition counts. Sepsis, AF/flutter, and heart failure meet the scale gate but require complete reruns because their smaller evaluated constructions were superseded. K is the number of discrete action classes.

Cohort	Subjects	Episodes	Action contract	Components / role	Current evidence status
Sepsis	22,437	27,236	Fluid $\times$ vasopressor, $K = 25$	P/R/T; aligned scaffold	Large-lineage rerun required
Respiratory support	18,709	21,679	PEEP $\times$ FiO ₂ , $K = 25$	P/R/T; policy extension	Current primary evidence
Shock	37,015	49,495	Fluid $\times$ vasopressor, $K = 25$	P/R/T; policy extension	Current primary evidence
AF/flutter	11,820	14,580	Rate $\times$ rhythm, $K = 25$	P/T; component-only	Large-lineage rerun required
AKI	16,380	21,083	P/T $K = 4$; constructed RRT $K = 2$	P/T; decision extension	Current primary evidence
Heart failure	27,611	32,552	P/T $K = 2$; constructed decision $K = 8$	P/T; decision extension	Large-lineage rerun required

Fidelity labels distinguish independent reimplementations from conceptual adapters. Deterministic controls are prespecified ϵ -soft distributions over the supported set with $\epsilon = 0.001$ for retrospective probability diagnostics; they are not scored by retrospective OPE.

4.3 Heterogeneous known-value and OPE evaluation

The primary policy surface crosses respiratory, shock, AKI-RRT, and heart-failure task profiles with six exact finite mechanisms: null response, a retained monotone control, interior optimum, state-dependent optimum, heterogeneous response, and delayed trade-off. Action counts, horizons, supported sets, behavior concentration, missingness, and termination prevalence are calibrated from aggregate EHR contracts; state dynamics, action response, toxicity, subtype, and counterfactual truth are specified synthetically. Dynamic programming supplies exact policy values. Hard gates require null-response equality, nonnegative exact regret, support compliance, categorical-CEM and receding-horizon execution, common-random-number precision, and complete inventory before method comparisons are read.

The four new mechanisms remove the fixed-extreme dominance built into the historical monotone control. They encode, respectively, an intermediate dose optimum, an optimum that varies with observed state, partially observed response heterogeneity, and delayed benefit versus immediate cost. An earlier composite adaptive family, in which a prespecified severity rule nearly recovers the exact policy, remains a separate construction sensitivity. Historical monotone matrices are retained only for evaluator regression testing.

All action responses, rewards, and termination mechanisms are known by construction. World-model architectures and policy methods are retrained within each environment; EHR-trained weights are not evaluated in these simulators. Five fixed training seeds compare five controls, five model-free methods, 24 world-model-planner combinations, and the exact dynamic-programming reference. Fixed-action controls are distinct from the state-dependent severity rule.

Exact dynamic-programming value is the primary estimand for every frozen policy. A separate common-random-number Monte Carlo check evaluates all policies from the same initial-state and exogenous-noise streams; policy randomness, training seeds, and evaluation seeds are separately recorded. Exact/Monte Carlo agreement and paired behavior-difference precision are hard evaluator checks, not substitutes for exact value. The complete policy-level

release retains exact return, regret, behavior delta, Monte Carlo return and standard error, unsupported mass, training seed, and model-exploitation gap.

The OPE matrix tests trajectory IS and WIS; per-decision IS, WPDIS, and CWPDIS; DR and WDR; FQE; and support-restricted variants [7, 11, 26]. A contract fixes estimator, behavior denominator, clipping, support rule, horizon, and mechanism. For each of the four heterogeneous mechanisms, 200 independent logged datasets are generated for each current respiratory and shock profile and 500 for AKI-RRT. Six frozen target policies span behavior, random and fixed supported controls, a severity rule, and the exact dynamic-programming policy. Behavior denominators and nuisance models are refit inside each dataset, and each interval uses only that dataset. Empirical coverage is the fraction of independently generated datasets whose interval contains the exact policy value; a 95% Wilson interval reports its sampling precision. Approval requires minimum policy-specific empirical coverage ≥ 0.90 ; Spearman rank and pairwise-order recovery ≥ 0.80 ; behavior-relative sign recovery ≥ 0.90 ; false-improvement probability ≤ 0.05 ; median ESS $\geq \max(100, 0.05N)$; zero unsupported target mass; and finite required values. The former single-dataset policy-set interval inclusion rate is retained as diagnostic-only and is not an authorization gate. Null and composite-adaptive repeated-dataset grids remain secondary evaluator sensitivities.

5 RESULTS

5.1 RQ1: P/R/T component fidelity

The complete current primary transition inventory comprises persistence/LOCF and HGB controls plus four P/R/T component families on respiratory, shock, and AKI. Sepsis, AF/flutter, and heart-failure rows from their smaller constructions are retained in the historical ledger but excluded from every primary aggregate until the large-lineage reruns finish. Among the three current cohorts, the Transformer has the lowest one-step RMSE in 3/3, whereas the Gaussian ensemble has the lowest final-horizon RMSE in 3/3. Thus one-step selection reverses on every currently qualified task. These are descriptive point leaders: neural methods have three seeds, the derived ensemble has one task-level row, and no subject-cluster leader interval is available.

Table 3 consolidates every executed comparison layer. It shows a stronger conclusion than a single aggregate leaderboard: some methods dominate one frozen metric, but the identity of the leader changes across point prediction, probabilistic recursion, adaptive policy comparison, and evaluator authorization. The

complete layer-specific inventory and every granular result row are released.

Table 4 reports every transition method under the same task-balanced aggregation rather than only winners. Persistence/LOCF has no probabilistic output, and HGB has no matched action-information row; these missing cells are structural rather than suppressed results. The current primary table contains all 18 method-cohort rows from the three unchanged lineages; the historical release also preserves the 18 superseded small-lineage rows with an explicit status flag.

Under the fixed action-information estimand, the current cohort means are 0.00228 for respiratory, 0.00219 for shock, and 0.00084 for AKI. Episode-cluster intervals exclude zero in all 10 model-seed rows in each cohort. Statistical separation can coexist with negligible magnitude; this diagnostic measures factual predictive information, not an action effect or policy value. Termination comparisons from the superseded small sepsis construction are not carried into the primary result.

Uncertainty is also metric-dependent. Across HGB and the four probabilistic component families, the Transformer has the lowest one-step CRPS in all three current cohorts. Under recursive roll-out, the ensemble has the lowest CRPS in 3/3, whereas GRU-D has the lowest MACE in 3/3. Recursive 90% coverage across methods ranges from 0.768 to 0.935 in respiratory, 0.765 to 0.926 in shock, and 0.728 to 0.943 in AKI. Hence neither one-step point error nor one-step uncertainty identifies recursive sharpness and calibration jointly.

5.2 RQ2: Heterogeneous known-value policy performance

All 18 current scale-qualified task-mechanism environments and 3,060 trained policy instances pass exact/Monte Carlo agreement and support checks; the current release includes 2,160 component-model-planner cells and an empty execution-failure ledger. The frozen discrimination check asks whether any learned policy-seed row exceeds both matched minimum- and maximum-action controls. It passes in 8 of 12 prespecified new task-mechanism cells: 3/4 each for respiratory and shock and 2/4 for AKI-RRT. Delayed trade-off remains fixed-control-led in all three profiles. Among the eight successful cells, the best recorded learned row is model-free in five and model-based in three. This cell-maximum check establishes that the mechanism family is not uniformly extreme-action dominated; it is not training-seed inference or a universal method ranking.

Figure 1 puts every evaluated family into one readable map rather than displaying only behavior-policy rows or winner labels. Panels A and B contain all transition and uncertainty baselines; Panels C and D contain all five controls, five model-free methods, and 24 component-model-planner labels. Poor and strong baselines coexist, and the leading family changes by task, response mechanism, horizon, and metric. All cohort-specific and policy-specific rows remain in the artifact rather than being filtered to winners.

5.3 RQ3: Exploitation and OPE disposition

The prior K25-sepsis policy diagnostic belongs to the superseded 3,440-episode construction and is therefore historical evidence only. It is not used to accept or reject the restored 27,236-episode lineage. The preregistered large-lineage sequence reruns the complete probability, support, collapse, and seed-stability gate before any sepsis policy statement is restored.

Component-model planners also exhibit substantial simulator-to-truth disagreement. Across the 2,160 current planner-seed cells, task-level mean absolute learned-model-versus-exact-return gaps range from 0.060 to 1.185, with a maximum of 1.826. These values remain on task-specific constructed reward scales and are not compared as cross-task performance; they show why a planner's value inside its learned model cannot replace exact known truth.

Repeated-dataset evaluation generates 3,600 independent logged datasets and evaluates 2,592 current frozen OPE-contract tuples over all four heterogeneous mechanisms. Table 5 shows the result: no tuple passes all gates. Coverage passes for only 10 tuples, rank for 309, pairwise order for 407, behavior-relative sign for 58, false improvement for 2,407, and ESS for 648; support and finiteness pass for all 2,592. This is a joint-gate failure decomposition rather than evidence that every estimator is universally unusable. The former fixed-dataset interval-inclusion result is retained as a *policy-set interval inclusion rate* in the historical artifact, and the null/composite-adaptive repeated grid remains a secondary sanity sensitivity. No estimator is transported by family name and no retrospective EHR OPE value is reported.

5.4 RQ4: Cross-cohort evaluability

All six frozen target lineages exceed 10,000 subjects and 10,000 episodes, but higher evaluation layers differ. Respiratory and shock retain rich repeated actions, post-action reward proxies, P/R/T components, and matched known-value task profiles. AKI retains a P/T transition surface and a separate daily wait-versus-RRT constructed decision profile. AF/flutter remains component-only unless strict action-reward ordering succeeds on its restored lineage. Sepsis, AF/flutter, and heart failure cannot re-enter the final six-cohort numeric surface until the large-lineage P/R/T, uncertainty, action-information, and applicable policy diagnostics are recomputed. None of the current heterogeneous repeated-dataset OPE contracts authorizes retrospective scoring.

6 PREDICTION-TO-POLICY FAILURE DECOMPOSITION

Prediction quality is neither ignored nor treated as policy performance. An 80-row machine-readable contract matrix states which known-value properties come from EHR aggregates—action count/support, horizon, missingness, termination prevalence, and behavior concentration—and which are specified—state dynamics, action response, toxicity, subtype, and exact truth. World-model architectures are retrained independently in the known-value environments; EHR weights are not reused. Respiratory and shock form the aligned bridge, while AKI and heart failure are explicitly labeled action-contract mismatch diagnostics.

Table 3: End-to-end baseline and method atlas. CRPS, mean absolute coverage error (MACE), and exact regret are minimized. Point leaders are descriptive within a frozen layer and do not define a global cross-layer winner.

Layer	Metric	Evaluated inventory	Scope	Point-leader pattern
Transition	One-step RMSE	LOCF, HGB, GRU-D, Transformer, RSSM, ensemble (6)	3 current EHR cohorts	Transformer 3/3
Transition	Final-horizon RMSE	Same six methods	3 cohorts; step 11 (44 h)	Ensemble 3/3
Uncertainty	One-step CRPS / MACE	HGB plus four probabilistic components (5); LOCF unavailable	3 current EHR cohorts	CRPS: Transformer 3/3
Uncertainty	Recursive CRPS / MACE	Same five probabilistic methods	3 current EHR cohorts	CRPS: ensemble 3/3; MACE: GRU-D 3/3
Policy	Exact return / regret	5 controls + 5 model-free + 24 component-planner labels (34)	18 exact-finite environments; 3,060 seed rows	Learned > both extremes in 8/12 new cells; no universal winner
OPE	Repeated-dataset authorization	IS, WIS, PDIS, WPDIS, CWPDIS, DR, WDR, FQE, support-restricted WPDIS (9)	3,600 independent datasets; 2,592 tuples; 4 mechanisms	No joint pass; none retrospective

Table 4: Complete current task-balanced transition/component performance. Each value first averages executed seeds within a cohort and method, then takes an unweighted mean over respiratory, shock, and AKI. RMSE, CRPS, and MACE are minimized; Cov90 is empirical recursive 90% interval coverage. Δ_a is matched-shuffle minus observed-action RMSE on reward-relevant targets and is predictive, not causal. The three restored large lineages will be added only after full reruns.

Group	Method	1-step RMSE	Recursive RMSE	1-step CRPS	Recursive CRPS	Rec. Cov90	Rec. MACE	Δ_a
Control	Persistence/LOCF	0.795	1.163	–	–	–	–	–
Control	HGB residual	0.704	1.025	0.359	0.480	0.834	0.052	–
P/R/T	GRU-D	0.694	1.112	0.343	0.490	0.901	0.023	0.0014
P/R/T	Transformer	0.689	2.270	0.340	0.739	0.754	0.123	0.0012
P/R/T	Categorical RSSM	0.755	1.262	0.374	0.552	0.779	0.104	0.0030
P/R/T	Gaussian ensemble	0.690	0.969	0.341	0.462	0.934	0.054	0.0010

Table 5: Repeated-dataset OPE gate pass counts for the current scale-qualified subset. Each heterogeneous mechanism has 648 frozen contract tuples over six fixed target policies. Coverage is empirical across independent logged datasets; all thresholds are frozen in Methods.

Gate	Interior	State-dep.	Heterog.	Delayed
Empirical coverage	0	0	0	10
Rank	0	24	0	285
Pairwise order	0	30	4	373
Improvement sign	0	16	24	18
False improvement	584	600	575	648
ESS	126	216	216	90
Support	648	648	648	648
Finite values	648	648	648	648
All gates	0	0	0	0

The bridge is mixed rather than a stable prediction-to-policy law. In the aligned respiratory/shock subset, recursive RMSE versus exact policy regret has Spearman $\rho = 0.190$ ($n = 8$), uncertainty-calibration error versus planner seed dispersion has $\rho = 0.257$ ($n = 6$), and factual-action information versus policy discrimination has $\rho = 0.548$ ($n = 8$). Within-task family-rank concordance is -0.400 for respiratory and 1.000 for shock. These descriptive coefficients have small, dependent task-family samples and no confirmatory p-values. Together with the three-current-cohort one-step-to-recursive ranking reversal and heterogeneous exploitation gaps, they show that no single forecasting metric consistently predicts decision performance.

The resulting benchmark logic is compositional: a row may remain valid for P/R/T forecasting while failing planning discrimination, and a policy may remain valid for support/collapse analysis while failing value authorization. This prevents an upper-layer

failure from erasing lower-layer evidence or a lower-layer success from silently authorizing a higher-layer claim.

7 REPRODUCIBILITY AND ACCESS

The reader package follows dataset-, model-, and benchmark-card documentation principles [5, 19, 22]. It includes task, feature, action, reward, planner, model, and estimator cards; sanitized policy-level and tuple-level tables; a number-to-source audit; immutable source hashes; environment manifest; build commands; and credentialed reconstruction instructions. The current primary view exposes all 18 method-cohort transition rows, all 3,060 heterogeneous policy-seed rows, all 2,160 component-model-planner cells, horizon-resolved transition results, cohort-specific uncertainty summaries, and bridge coefficients. The complete historical ledger remains available but marks the 18 small-lineage transition rows and affected policy evidence as superseded rather than silently mixing them with the primary view. Code, contracts, synthetic fixtures, aggregate evidence, and clean-room commands are provided in the anonymous artifact at <https://anonymous.4open.science/r/ehrdyn-icu-65FB>. The source data are MIMIC-IV v3.1 [12]; access remains governed by PhysioNet credentialing and its data-use agreement. Raw EHR rows, identifiers, exact timestamps, trajectories, role-membership files, row-level predictions, tensors, and checkpoints are not redistributed.

Authorized users rebuild task arrays locally, fit all data-dependent transforms on development training roles, run development validation, execute exact known-value and repeated-dataset OPE evaluation, and regenerate the aggregate synthesis and PDF. Unrestricted users can validate the packaged contracts and aggregate hashes and run the known-value tests without MIMIC-IV. Independent credentialed reconstruction remains pending: it is a

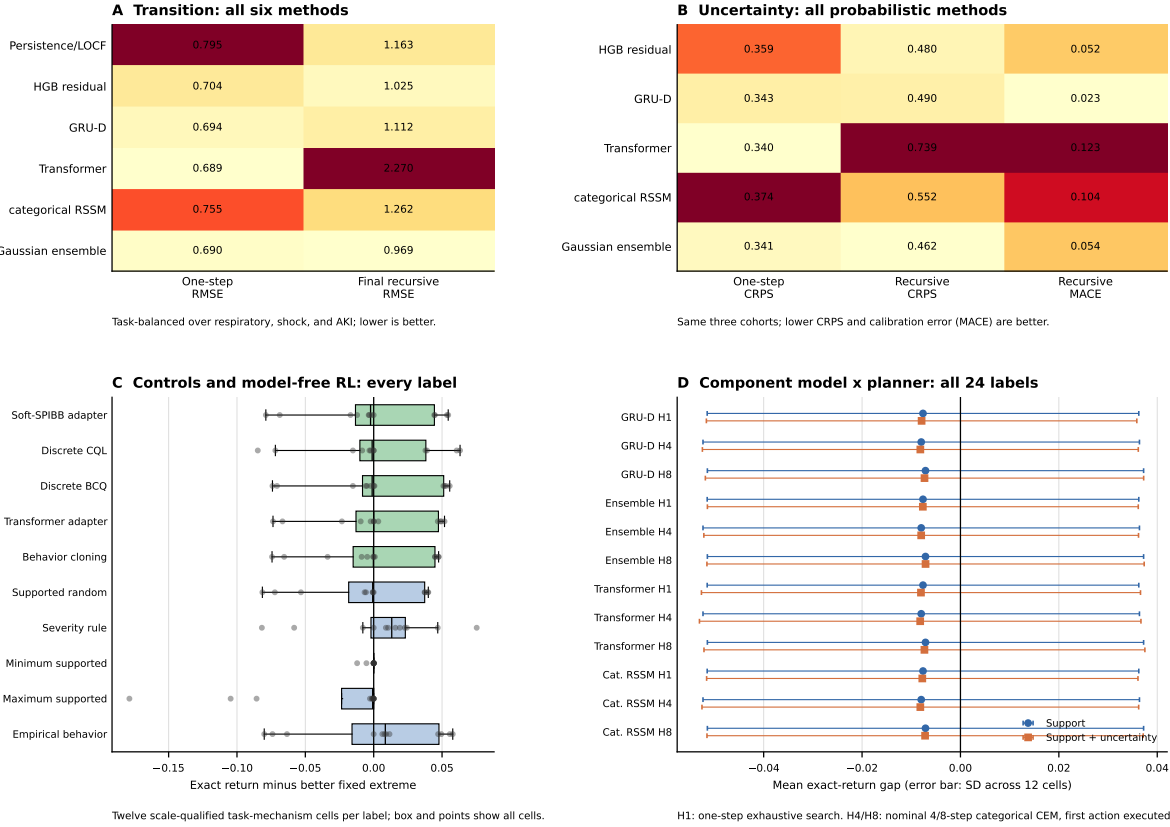
Complete current baseline atlas (large-lineage reruns pending for sepsis, AF/flutter, and heart failure)

Figure 1: All-layer benchmark atlas on the three current scale-qualified lineages. (A) All six transition methods, with task-balanced one-step and final recursive RMSE. (B) All five uncertainty-capable methods, with one-step and recursive CRPS/MACE. (C) Exact return minus the better fixed extreme for all five controls and five model-free methods across 12 new task-mechanism cells. (D) The same gap for all 24 P/R/T component-model-planner labels. H1 is exhaustive one-step supported-action search; H4 and H8 are nominal four- and eight-step categorical-CEM planners that execute the first action and replan; +U adds the uncertainty penalty. Bars/points summarize all current cells, not only winners. Sepsis, AF/flutter, and heart failure enter this figure only after their large-lineage reruns.

reproducibility check in which another MIMIC-credentialed user regenerates aggregate results from a clean clone, not external-data validation or a generalization experiment.

8 LIMITATIONS, DATA GOVERNANCE, AND SCOPE

The EHR findings are development benchmark evidence: no genuinely untouched confirmatory role is available. Component-model leader comparisons lack subject-cluster intervals, and the derived Gaussian ensemble contributes one row per task. Recursive evaluation predicts values under a fixed missingness convention rather than modeling a generative observation process; accordingly, we claim P/R/T components rather than a complete EHR world model with an observation head. The four-hour action is a recorded interval exposure, not guaranteed to be assignable at the bin start. Sepsis is an AI-Clinician-aligned K25 scaffold rather than an exact reproduction, and its reference terminal

outcome is unavailable. Restoring the large sepsis, AF/flutter, and heart-failure lineages is a submission-critical reevaluation: until it is complete, their old small-lineage metrics remain excluded and the numerical primary synthesis contains only respiratory, shock, and AKI. AKI and heart-failure policy comparisons use separately constructed known-value tasks rather than the original EHR transition actions, and AF remains component-only.

The policy mechanisms are exact and heterogeneous but constructed; their results do not establish value on retrospective EHR trajectories. The frozen 8/12 discrimination statistic is a maximum over the complete learned policy-seed cell and therefore does not support learned-family superiority. The heterogeneous repeated-dataset OPE audit covers all four new mechanisms but evaluates six fixed target policies rather than the entire trained-policy inventory; it authorizes no retrospective scoring. MIMIC-IV is restricted,

so the artifact shares code, contracts, synthetic fixtures, and aggregate results rather than row-level data. These are evidence boundaries of the technical benchmark.

9 CONCLUSION

We present a common decision-evaluation scaffold that connects EHR P/R/T component learning, recursive uncertainty, planning, model-free RL, heterogeneous known-value truth, and repeated-dataset OPE authorization. Across the current scale-qualified evidence, one-step, recursive, probabilistic, policy, and estimator layers produce different rankings and failure modes rather than a universal winner. By publishing exact contracts, every method row, fixed controls, supersession status, and non-execution rules, the benchmark makes those differences reusable while preventing small-lineage results from being mistaken for final evidence. Retrospective EHR policy ranking remains unavailable until an adaptive evaluator contract and policy-specific EHR gates both pass.

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